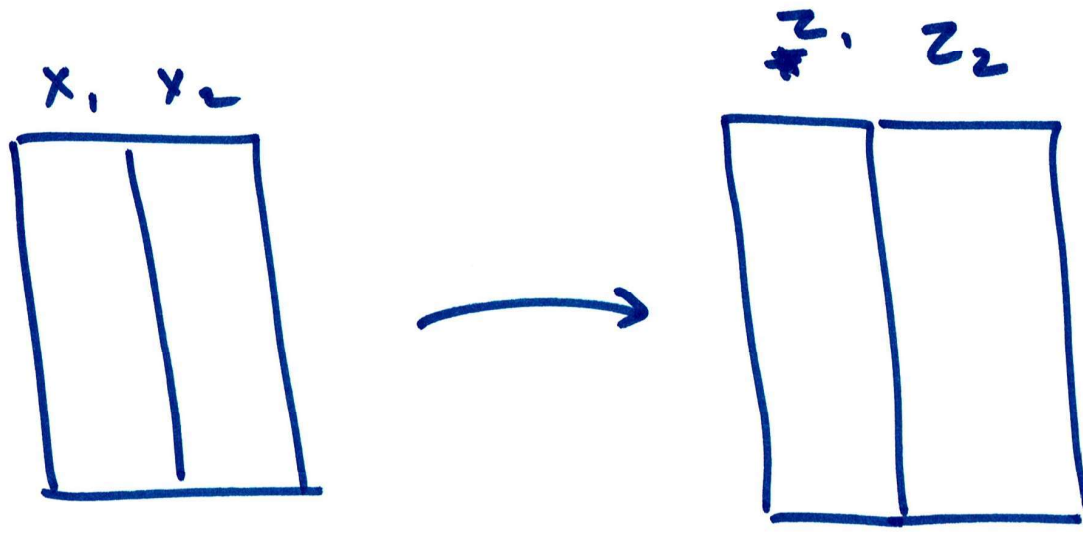
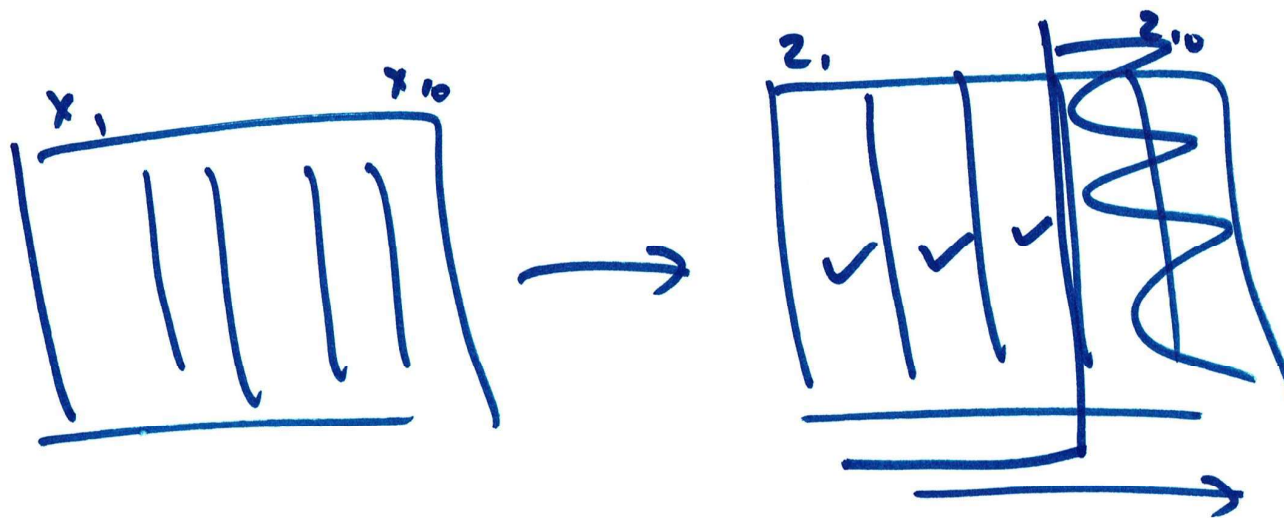
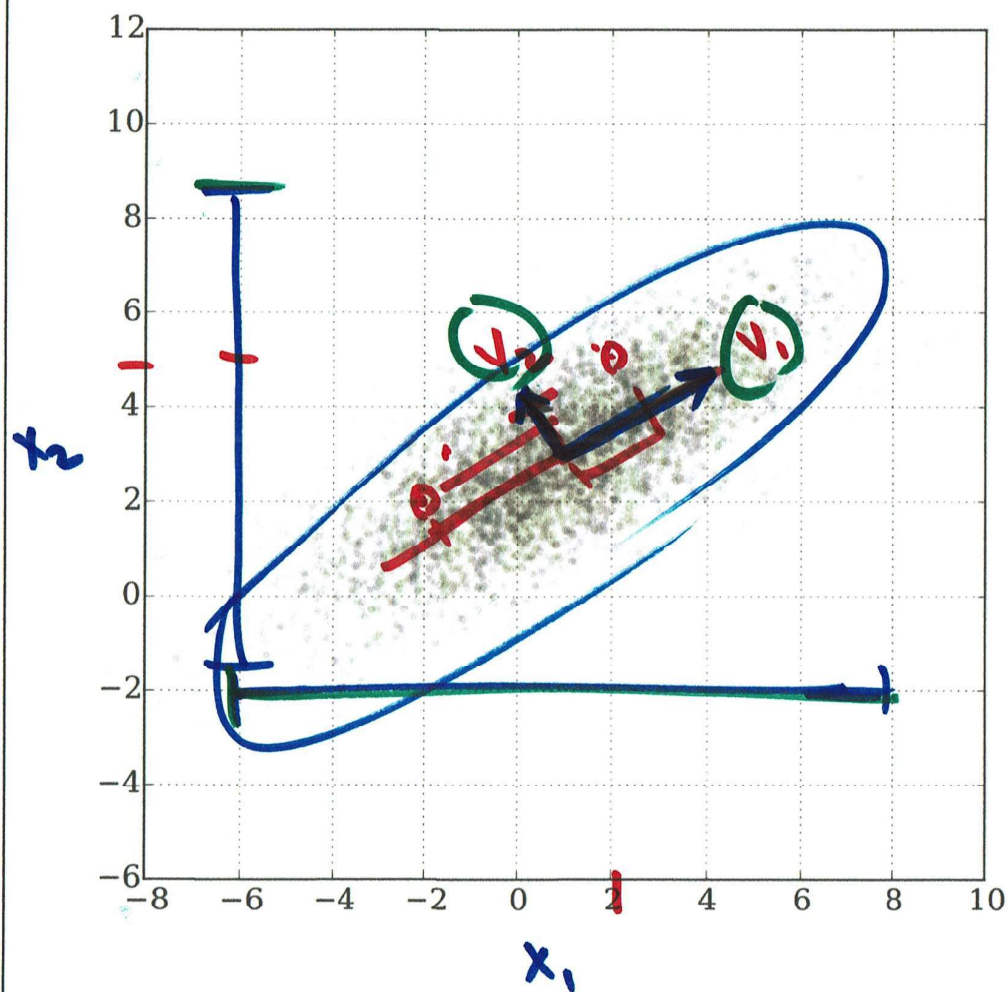








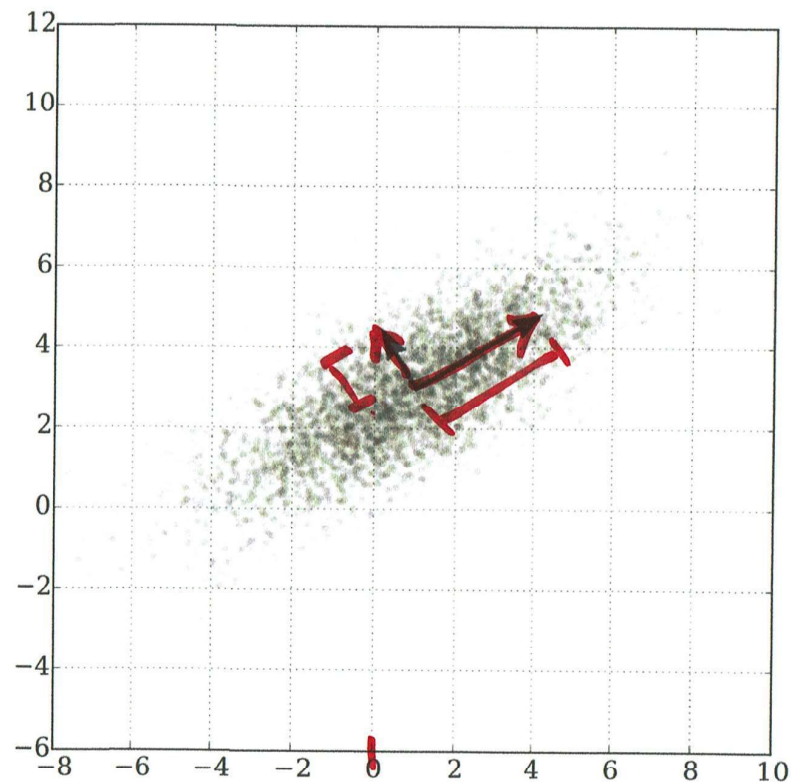
$$\begin{array}{c} \left\{ \begin{array}{c|c} x_1 & x_2 \\ \hline 2 & 5 \\ -2 & 2 \end{array} \right\} \rightarrow \left\{ \begin{array}{c|c} z_1 & z_2 \\ \hline 1 & 0.8 \\ -2 & 0.5 \end{array} \right\} \\ \text{1.2} \quad 0.9 \qquad \qquad \text{1.9} \quad 0.1 \end{array}$$

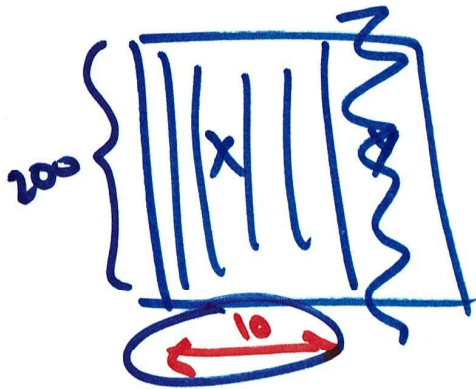
A diagram illustrating the principal components Z_1 and Z_2 . A horizontal red line represents Z_1 and a vertical red line represents Z_2 . Data points are projected onto these lines. Below the diagram, the equations for the principal components are given:

$$\begin{cases} Z_1 = \boxed{1.2} x_1 + \boxed{0.9} x_2 \\ Z_2 = \boxed{1.9} x_1 + \boxed{0.1} x_2 \end{cases}$$

PCA

- creates new variables using linear combinations of old variables
- is designed to create variables that are independent of one another
- also manages to tell us how important each of these new variables are
- this “importance”, helps us to choose how many variables we will use





$$X_{\text{new}} = \frac{X - \mu}{\sigma}$$

$$C_{\text{cov}} = \text{Cov}(X_{\text{new}})$$

Eigen decomposition

$$\begin{array}{c} \rightarrow \begin{array}{|c|c|c|c|} \hline e_1 & e_2 & \dots & e_{10} \\ \hline v_1 & v_2 & \dots & v_{10} \\ \hline \end{array} \end{array}$$

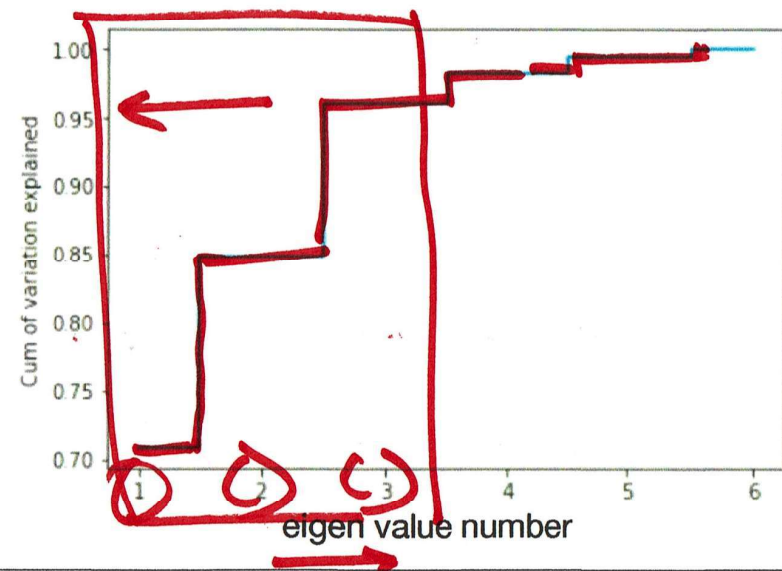
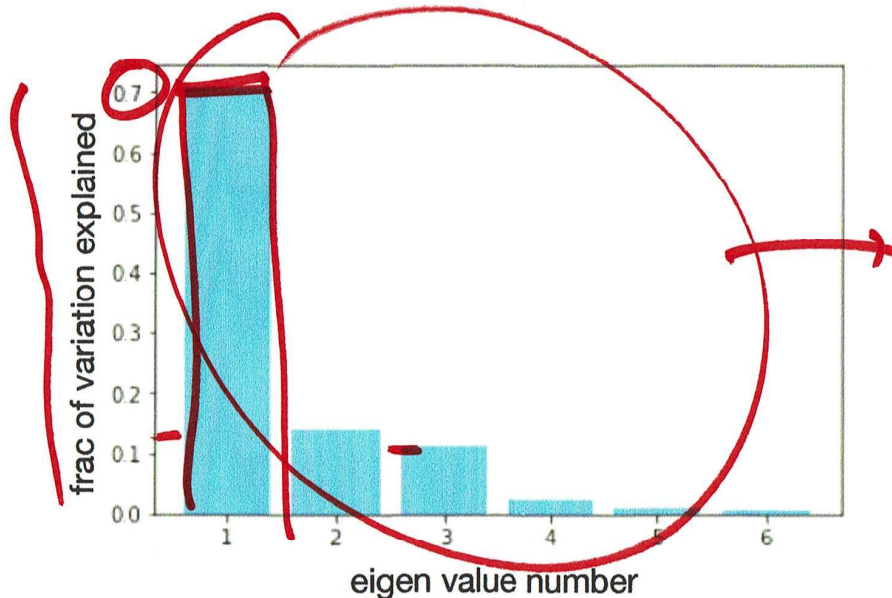
$$Z$$

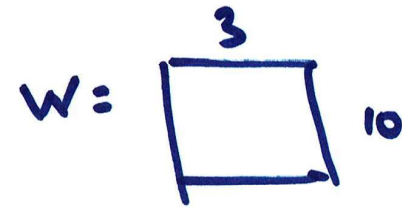
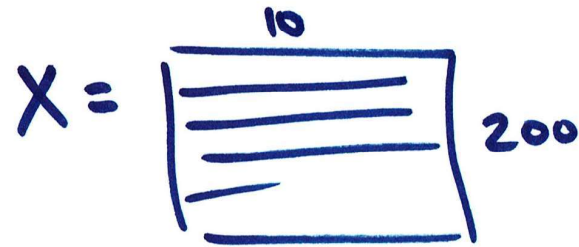
$$10 \times 3$$

$$W = \begin{pmatrix} v_1 & v_2 & v_3 \end{pmatrix}$$

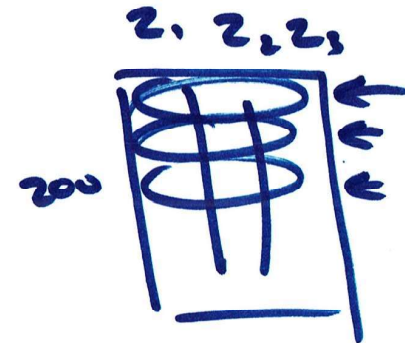
- Scale the data and compute the covariance matrix
- Break the covariance matrix into magnitude and direction. Eigen Vectors and the Eigen Values of the covariance matrix can be thought of as the natural axis/directions and magnitudes along those axis, of the data
- The eigen values also can be used to calculate the percentage of variation explained by each component
- Sort in the eigen values in descending order and calculate the cumulative percentage of variation explained
- Pick the number of principal components you will use
- Transform to new variables

$$\frac{e_1}{\sum e_i}, \frac{e_2}{\sum e_i}, \frac{e_3}{\sum e_i}$$





$\Rightarrow$ Z 200×3



$$S = 100 + 20 \underline{z_1} + 15 z_2 - 10 z_3$$

$$\begin{aligned} z_1 &= \square x_1 + \square x_2 + \square x_3 \\ z_2 &= \dots \\ z_3 &= \dots \end{aligned}$$

1... x₁₀